SOLPS5.0 & MATLAB

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- Purpose
- From b2fplasma to *.mat
- From a b2plot output to *.mat
- Postprocessing routines for the matfile
- Interesting matlab routines for JET Data
- Work in progress
- Future Work - Dreams

Purpose

- Started because at CRPP everything is done in matlab
- At JET and CRPP matlab 6.0 is available - powerful
- Experimental data accessible via matlab routines at JET and TCV - also via remote access through mdsplus
- "Quick" way to compare results from different runs - data base
- More direct comparison of experimental data and simulations
- Possibility of easy way to "manipulate" experimental and simulated data at same time; e.g. mapping to midplane

From b2fplasma to *.mat (I)

- SOLPS5.0 (B2.5-Eirene) saves files as unformatted b2fplasma (B2.5) and fort44 (Eirene)
- Transform unformatted b2fplasma into formatted b2fplasmf, e.g.:

/home/mwisch/solps5.0_eirene/src/Braams/b2/runs/D+C+He/Jet/8MW/rundirectory

do gmake -f ../../../../Makefile b2uf.prt

in matlab do:

```
>>!cd /home/mwisch/solps5.0_eirene/src/Braams/b2/runs/D+C+He/Jet/8MW/rundirectory
>> /home/mwisch/matlab/solps5/solps5_b2fplasmaf;see help read_b2fplasmaf
```

From b2fplasma to *.mat (II)

- Filename: shotnumber_serialnumber.mat; e.g. 54002_13.mat; may vary as user wishes
- Saved as matlab structure:

solps5

  .b2 ; contains variables stored in b2fplasma - B2.5 output

e.g. solps5.b2.ne: electron density

  .eirene ; contains variables stored in fort44 - Eirene output

e.g. solps5.b2.dab2: atomic densities

- description of stored variables - as far as I know them, and as far as I understood the structure of their matrix - soon available under public web page
- check: crppwww.epfl.ch/~wischmeier from time to time!!

From a b2plot output to a *.mat

- Currently only for b2plot output on b2-grid; not possible yet for written output along diagnostic chords -> will be done soon because of urgent need!
- Use the mprt command for b2plot to create an ascii output; e.g.: echo “ ´bolo.cmd´ cmd pvol ´bolo.dat´ mprt ¦ b2plot
- in matlab do:

\>>/home/mwisch/matlab/solps5/diagnostic/
mw_read_b2plot_mprt
Look also at help mw_read_b2plot_mprt for further info

Postprocessing routines for the mat-files

- Diagnostic routines which use the file created with read_b2fplasmaf
- Loading a file: mw_jet_loadsolps5fun.m; allows to create an array of solps5 files; e.g solps5(1), solps5(2) etc.

=> build database of runs which you want to compare

- Routines that postprocess solps5 data:
- mw_heI_radiationI; HeI 728nm and HeI 668nm
- mw_balmer_s5; H-alpha & H-beta

both use amjuel rates from CR model (H.12)

- amjuel.mat is AMJUEL file in matlab format

e.g.: rates.amjuel.h4.r2_1_5

- mw_rates_plotting plots amjuel rates dependent on ne and Te

Interesting matlab routines for JET data (I)

- **YOU WANT JET DATA!!!**
- Example: Routines developped by Dr. Olivier Sauter CRPP/ EPFL: *gdat*
- In your *startup.m* file add the following paths doing e.g.:

addpath(´/home/osauter/matlab/gdat´,´-end´)
addpath(´/home/osauter/matlab/JETmatlab´,´-end´)
gdatpaths

- Then set a global variable as follows for accessing MDSPLUS:

global usemdsplus
usemdsplus = 1;
NOW YOU CAN START USING THE ***GDAT*** *PACKAGE*

Interesting matlab routines for JET data (II)

- YOU want an EFIT file for DG / Carre:

efitdata = geteqdskJET(53981,60,33,65,´/tmp´);

Shotnumber

time grid directory

- Produces a file /tmp/EQDSK.53981t60.000000
- For plotting look at *help geteqdskJET;*
- Can now use: equtrn/e2d filename filename.equ for generating required input of DG - Carre

Interesting matlab routines for JET data (III)

- ***You want to access experimental data as e.g Langmuir probes or Li-beam profiles...***
- ***You know the names of the ppf´s, DDA´s, rdm´s,uid etc.***
- ***Use gdat.m; examples:***
- ***langmuir=gdat(nshot,´ppf/s32b/te?uid=jetrdm´,1,´JET´);***

Don´t need it if ppf without uid

plot it

Machine

- ***ky63=gdat(54002,´ppf/ky63/ner?uid=jetxmg´,1,´JET´);***
- Look also at help gdat
- Output is structured array: ky63 = data: [29x112 double]
  x: [29x1 double]
  t: [112x1 double]
- Data is then of the sort ky63.data or time as ky63.t

WORK IN PROGRESS

- AT CRPP:
- Jan Horacek working on a *psi-toolbox*, which will allow mapping from target to midplane at JET
- Myself: Read out of "b2plot chord written output" into matlab
- Routines as required while I analyse the simulations....
- If routine of general interest it will be put into /home/mwisch/matlab/solps5/diagnostic
- **At other places:**
- I don´t know....
- **FUTURE: DREAMING OF A GUI...**